

Digital Obstructions to Signed Pascal Representations of Lucas Numbers

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Abstract

Let L_n be the n th Lucas number. We study signed representations

$$L_n = \sum_{k=0}^n \varepsilon_k \binom{n}{k}, \quad \varepsilon_k \in \{-1, 0, 1\},$$

and write $a(n) = 0$ when no such representation exists. The zero terms of the resulting sequence lead to a local obstruction theory. Lucas' theorem collapses a hypothetical representation modulo a prime to a small system determined by the base- p digits of n , while Frobenius determines the corresponding Lucas target. Incompatible local systems attached to different sparse descriptions of the same index then certify global nonrepresentation. We prove the first nontrivial zero $a(26) = 0$, establish parametric families of zeros, and develop local signatures, range-exact lifting, nested digital descent, matching, a mod-5 color law, ramified behavior at 5, and saturation. Within a precisely defined low-dimensional certificate class we classify all primitive obstructions into four types. We then exhibit a genuinely different two-digit tensor-selector family outside that class; it explains the zeros 112, 113, 142 and gives a direct hand proof for the first known odd zero 113. An appendix contrasts the Lucas problem with a uniform signed residue-class construction for Fibonacci numbers.

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1 A399744 and the mystery of the zeros

Let $(L_n)_{n \geq 0}$ be the Lucas numbers,

$$L_0 = 2, \quad L_1 = 1, \quad L_n = L_{n-1} + L_{n-2}.$$

For $n \geq 0$, define $a(n)$ to be the minimum number of nonzero coefficients in a representation

$$L_n = \sum_{k=0}^n \varepsilon_k \binom{n}{k}, \quad \varepsilon_k \in \{-1, 0, 1\},$$

where equal binomial coefficients in different positions are regarded as distinct entries. If no such representation exists, set $a(n) = 0$. This is OEIS A399744 [4].

Classical identities connect Fibonacci and Lucas-type sequences with sums of binomial coefficients; see, for example, Gordon [2, 3] and the standard background in Koshy [6]. The problem considered here is different in two respects: each individual position of the Pascal row may be used only with coefficient -1 , 0 , or 1 , and the principal question is exact nonrepresentation rather than the evaluation of a prescribed binomial sum.

For many additive representation problems one expects sufficiently large rows to provide abundant flexibility. The appearance of exact nonrepresentation is therefore surprising: a zero here means that the entire signed coefficient box $\{-1, 0, 1\}^{n+1}$ misses the Lucas target.

A simple positive example is

$$L_{15} = 1364 = \binom{15}{4} - \binom{15}{0},$$

so $a(15) = 2$.

The first 75 values, for $0 \leq n \leq 74$, are 0, 1, 2, 2, 2, 2, 3, 3, 3, 3, 4, 4, 5, 5, 7, 2, 7, 6, 6, 6, 10, 9, 7, 7, 7, 9, 0, 11, 10, 11, 12, 9, 10, 12, 15, 10, 11, 11, 0, 15, 16, 14, 13, 14, 0, 16, 15, 15, 15, 18, 20, 17, 19, 18, 18, 18, 19, 21, 22, 20, 20, 19, 20, 20, 21, 25, 23, 19, 23, 23, 22, 23, 25, 22, 0.

Thus the nontrivial zero indices in this range are

$$26, 38, 44, 74.$$

The positive zero indices of A399744 form OEIS A399745 [5]. Since A399744 is completely determined through $n = 74$, the first four terms of A399745 are therefore

$$26, 38, 44, 74.$$

For empirical orientation, the companion exact enumeration has established the first eleven positive zero indices as

$$26, 38, 44, 74, 104, 110, 112, 113, 132, 142, 158.$$

The ordering of these later data is *not used in any proof below*. Nine of these eleven indices receive independent hand-proof mechanisms in the present paper; the two deliberately unresolved architectural cases are 110 and 132. A detailed index/mechanism/proof-status table is deferred to Section 10, after the proof architectures have been developed. The odd zero 113 is already conceptually important: every Type-I–IV realization is even, so it gives an external reason to expect a mechanism beyond the low-dimensional system rather than a retrospective enlargement of that system.

The zeros have a qualitatively different meaning from the positive terms. A positive value measures the shortest signed representation; a zero asserts that no signed representation exists. The central question is therefore:

Why can signed Pascal representations fail at all, and why at these special indices?

The answer developed below is local and digital. A sparse base- p expansion of n makes most coefficients in row n vanish modulo p . Simultaneously, $L_n \pmod{p}$ is controlled by a Frobenius action. If several sparse descriptions of the same n constrain the same few boundary signs incompatibly, the global representation cannot exist.

Figure 1 summarizes the mechanism used throughout the paper. Sections 2–7 develop the proof-producing obstruction theory. Section 8 then isolates a bounded low-dimensional certificate class and proves a completeness theorem within that class. Section 9 exhibits a reusable

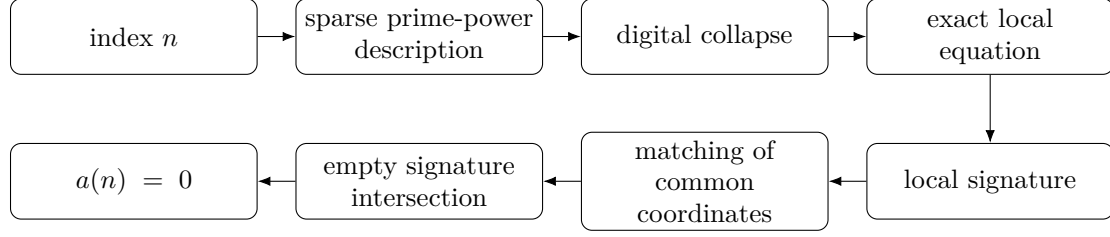


Figure 1: The proof-producing pipeline. Sparse prime descriptions collapse a large Pascal row to exact local equations; signatures are compared only when they refer to the same global coordinates. An empty matched intersection certifies nonrepresentation.

mechanism outside it. The explicit zero proofs and parametric families do not depend on the classification theorem.

The mechanism labels are a roadmap, not a global taxonomy. Types I–IV are the four primitive obstructions in the low-dimensional class of Definition 8.3; Type V is shorthand for the first reusable architecture proved outside that class. The distinction is summarized here:

Type	Status	First realization	Informal description
I	low-dimensional class	38	incompatible endpoint value
II	low-dimensional class	44	incompatible boundary regions
III	low-dimensional class	26	a ramified selector chooses a point later excluded
IV	low-dimensional class	31250	the last regular intersection point is removed by a ramified endpoint set
V	outside class	112, 113	a two-digit tensor collapse excludes the surviving local selector data

The index 74 realizes Types I and II simultaneously. The first three classified architectures already occur for $n \leq 74$. Type IV is much rarer: Section 6.3 proves its first realization in the class at $31250 = 2 \cdot 5^6$. Type V is introduced only after Theorem 8.8; the table is a roadmap for the paper, not a five-type completeness claim.

1.1 The first mystery: $n = 26$

Before introducing the general theory, we exhibit the phenomenon in full.

Proposition 1.1. $a(26) = 0$.

Proof. Assume that

$$L_{26} = \sum_{k=0}^{26} \varepsilon_k \binom{26}{k}.$$

Put

$$A = \varepsilon_0 + \varepsilon_{26}, \quad B = \varepsilon_1 + \varepsilon_{25}, \quad C = \varepsilon_2 + \varepsilon_{24}, \quad D_3 = \varepsilon_3 + \varepsilon_{23}.$$

First use $26 = 2 \cdot 13$. Modulo 13, Lucas' theorem leaves only positions 0, 13, 26 with weights 1, 2, 1, while $L_{26} \equiv L_2 = 3 \pmod{13}$. Hence

$$A + 2\varepsilon_{13} \equiv 3 \pmod{13}.$$

The left side lies in $[-4, 4]$, so equality holds in $\mathbb{Z}$, and the only possibility is

$$A = 1, \quad \varepsilon_{13} = 1. \quad (1.1)$$

Next use $26 = (101)_5$. Modulo 5, only positions 0, 1, 25, 26 survive, all with weight 1, and $L_{26} \equiv 3 \pmod{5}$. Thus

$$A + B \equiv 3 \pmod{5}.$$

Since $A + B \in [-4, 4]$, it is 3 or -2 . With $A = 1$, the second alternative would give $B = -3$, impossible. Hence

$$B = 2. \quad (1.2)$$

Finally, $26 = 23 + 3$ and $\left(\frac{5}{23}\right) = -1$. Modulo 23 the surviving row is $(1, 3, 3, 1)$ and the Lucas target is $-L_2 = -3$. Since $23 > 16 + 3$, the congruence lifts to

$$A + 3B + 3C + D_3 = -3.$$

Using (1.1)–(1.2) gives $3C + D_3 = -10$, whereas $C, D_3 \in [-2, 2]$ imply $3C + D_3 \geq -8$. Contradiction. $\square$

The proof illustrates the basic mechanism in its simplest form: the reductions modulo 13 and 5 successively force $A = 1$ and $B = 2$, while the reduction modulo 23 excludes those boundary data. No single global signed representation can satisfy all three local conditions.

A logically independent comparison. The obstruction theory begins in the next section. For perspective only, Appendix A derives a uniform signed two-residue construction for Fibonacci numbers and contrasts it with the analogous Lucas residue pattern, whose symmetry-fixed residue class carries the forbidden coefficient 2. Nothing in Sections 2–10 depends on that comparison.

2 Digital collapse

Two simplifications occur at the same time. **Lucas' theorem [7, 1] collapses the Pascal side:** most binomial coefficients disappear modulo p , leaving a small row determined by the base- p digits of n . **Frobenius collapses the Lucas side:** the target L_n becomes a much smaller Lucas number, with a sign determined by whether Frobenius fixes or exchanges the two roots of $x^2 - x - 1$.

Example: 26 modulo 13. In base 13 one has $26 = (20)_{13}$. Lucas' theorem allows only digit choices $(00), (10), (20)$, hence only the positions 0, 13, 26 survive. Their weights are $\binom{2}{0}, \binom{2}{1}, \binom{2}{2} = (1, 2, 1)$. Thus the huge row 26 collapses modulo 13 to the small Pascal row $(1, 2, 1)$. On the target side Frobenius is split, so $L_{26} \equiv L_2 = 3 \pmod{13}$.

Example: 26 modulo 5. Now $26 = (101)_5$. Lucas' theorem allows the digit choices $(000), (001), (100), (101)$, corresponding to the positions 0, 1, 25, 26. All four surviving binomial weights are 1, so the Pascal side collapses to

$$\varepsilon_0 + \varepsilon_1 + \varepsilon_{25} + \varepsilon_{26} = A + B.$$

This example shows that a sparse expansion need not collapse to a single small Pascal row of the form $(1, 2, 1)$; several separated digit blocks may survive. The prime 5 is ramified for

$x^2 - x - 1$, so the split/nonsplit Frobenius dichotomy of Theorem 2.1 is not invoked here; its target behavior is treated separately in Section 6.

Let $p \neq 5$ be an odd prime and write

$$n = \sum_{i=0}^t a_i p^i, \quad 0 \leq a_i < p.$$

Define

$$s = \sum_i a_i, \quad E = \sum_{i \text{ even}} a_i, \quad O = \sum_{i \text{ odd}} a_i.$$

Theorem 2.1 (Digital collapse). *If*

$$L_n = \sum_{k=0}^n \varepsilon_k \binom{n}{k}, \quad \varepsilon_k \in \{-1, 0, 1\},$$

then

$$\sum_{0 \leq b_i \leq a_i} \left(\prod_i \binom{a_i}{b_i} \right) \varepsilon_{\sum_i b_i p^i} \equiv U_p(n) \pmod{p},$$

where

$$U_p(n) = L_s \quad \text{if } \left(\frac{5}{p}\right) = 1,$$

and

$$U_p(n) = (-1)^{\min(E,O)} L_{|E-O|} \quad \text{if } \left(\frac{5}{p}\right) = -1.$$

Proof. Lucas' theorem gives

$$\binom{n}{k} \equiv \prod_i \binom{a_i}{b_i} \pmod{p}$$

for $k = \sum_i b_i p^i$, and the product vanishes if some $b_i > a_i$.

For the target side, let α, β be the roots of $x^2 - x - 1$ in an algebraic closure of $\mathbb{F}_p$. Since $p \neq 5$, the roots are distinct, $L_m = \alpha^m + \beta^m$, and $\alpha\beta = -1$. Intuitively, Frobenius has only two possibilities: it either fixes the two roots or exchanges them. The distinction is governed by the discriminant 5 of $x^2 - x - 1$: if 5 is a square modulo p , the two roots already lie in $\mathbb{F}_p$ and Frobenius fixes them; if 5 is a nonsquare, the roots form a conjugate pair over $\mathbb{F}_{p^2}$ and Frobenius exchanges them.

If $\left(\frac{5}{p}\right) = 1$, Frobenius fixes both roots. Hence $\alpha^{p^i} = \alpha$ and $\beta^{p^i} = \beta$, so $\alpha^n = \alpha^s$ and $\beta^n = \beta^s$.

If $\left(\frac{5}{p}\right) = -1$, Frobenius exchanges the roots. Thus powers in even digit positions contribute to the same root and powers in odd digit positions to the other, giving

$$\alpha^n = \alpha^E \beta^O, \quad \beta^n = \beta^E \alpha^O.$$

Factoring $(\alpha\beta)^{\min(E,O)} = (-1)^{\min(E,O)}$ yields the stated formula.

For a small illustration of the minus mechanism, if a base- p pattern has one p -digit and three unit digits, then

$$L_{p+3} = \alpha^{p+3} + \beta^{p+3}.$$

In the nonsplit case $\alpha^p = \beta$ and $\beta^p = \alpha$, so

$$L_{p+3} = \alpha^3 \beta + \beta^3 \alpha = (\alpha\beta)(\alpha^2 + \beta^2) = -L_2.$$

This is the source of the negative local target. □

The local complexity is therefore digital rather than metric: in the split case only the digit sum matters on the Lucas side; in the nonsplit case one retains the even/odd position split.

3 Range-exact sparse collapses and the first regular families

The main applications start from an index with only two relevant base- p blocks.

Definition 3.1 (Range-exact lifting). Suppose an integer quantity S is known a priori to lie in $[-M, M]$ and satisfies

$$S \equiv T \pmod{p},$$

with $T \in [-M, M]$. We call the lift range-exact if T is the unique integer in $[-M, M]$ congruent to T modulo p . Equivalently,

$$p > M + |T|.$$

Corollary 3.2 (Exact small-row collapse). *Let*

$$n = p^u + r, \quad u \geq 1, \quad 1 \leq r < p,$$

with $p \neq 5$ an odd prime. Put

$$\sigma = \left(\frac{5}{p}\right)^u, \quad Y_j = \varepsilon_j + \varepsilon_{n-j}.$$

Then

$$\sum_{j=0}^r \binom{r}{j} Y_j \equiv F_r^\sigma \pmod{p},$$

where

$$F_r^+ = L_{r+1}, \quad F_r^- = -L_{r-1}.$$

The left side lies in $[-2^{r+1}, 2^{r+1}]$. Consequently the lift is range-exact precisely under

$$p > 2^{r+1} + L_{r+1} \quad (\sigma = +), \quad (3.1+)$$

$$p > 2^{r+1} + L_{r-1} \quad (\sigma = -). \quad (3.1-)$$

In particular the uniform bound $p > 2^{r+1} + L_{r+1}$ is always sufficient.

Proof. The base- p digits are r at position 0 and 1 at position u . The surviving indices are j and $p^u + j$ with weight $\binom{r}{j}$; reindexing the second half by $j \mapsto r - j$ gives Y_j . The target follows from Theorem 2.1. Finally,

$$\left| \sum_{j=0}^r \binom{r}{j} Y_j \right| \leq 2 \sum_{j=0}^r \binom{r}{j} = 2^{r+1},$$

and range-exactness gives the two thresholds. □

For $r = 1$ we obtain the endpoint lines

$$\sigma = +1 : A + B = 3, \quad \sigma = -1 : A + B = -2, \quad (3.2)$$

where $A = \varepsilon_0 + \varepsilon_n$ and $B = \varepsilon_1 + \varepsilon_{n-1}$. In particular the negative line forces

$$A \in \{-2, -1, 0\}. \quad (3.3)$$

3.1 The doubling atom

If $n = 2p^u$ with $p \geq 11$, Theorem 2.1 gives

$$A + 2\varepsilon_{p^u} \equiv 3 \pmod{p}.$$

The left side lies in $[-4, 4]$, hence

$$A = 1, \quad \varepsilon_{p^u} = 1. \tag{3.4}$$

We call $A = 1$ the *doubling atom* and denote its endpoint signature by $D = \{1\}$.

Theorem 3.3 (Prime-power doubling obstruction). *Let $p \geq 11$ be prime and $u \geq 1$. If*

$$q = 2p^u - 1$$

is prime and $\left(\frac{5}{q}\right) = -1$, then

$$a(2p^u) = 0.$$

Proof. Set $n = 2p^u = q + 1$. The base- p collapse gives $A = 1$. The base- q collapse with $r = 1$ and negative Frobenius sign gives $A + B = -2$, hence $A \leq 0$. Contradiction. $\square$

Remark 3.4 (A further Type-I zero). The same endpoint mechanism gives a short proof that 386 is another member of A399745. Indeed,

$$386 = 2 \cdot 193 = 379 + 7,$$

and 193 and 379 are prime. The doubling atom gives $A = 1$. Since $379 \equiv 4 \pmod{5}$, the direct terminal-7 collapse has positive sign and is range-exact because

$$379 > 2^8 + L_8 = 303.$$

Hence

$$\sum_{j=0}^7 \binom{7}{j} Y_j = L_8 = 47.$$

Reducing this exact equality modulo 7 gives $A + Y_7 \equiv 47 \equiv -2 \pmod{7}$. As $A + Y_7 \in [-4, 4]$, necessarily $A + Y_7 = -2$, so $A \leq 0$, contradicting $A = 1$. Thus $a(386) = 0$. We do not claim here that 386 is the fifth term of A399745.

3.2 A weighted two-dimensional obstruction

Theorem 3.5 (Prime-power gap-two obstruction). *Let $p > 23$ be prime with $p \equiv 1 \pmod{5}$, and let $u \geq 1$. If*

$$q = p^u + 2$$

is prime, then

$$a(p^u + 3) = 0.$$

Proof. Set $n = p^u + 3 = q + 1$ and

$$A = \varepsilon_0 + \varepsilon_n, \quad B = \varepsilon_1 + \varepsilon_{n-1}, \quad C = \varepsilon_2 + \varepsilon_{n-2}, \quad D_3 = \varepsilon_3 + \varepsilon_{n-3}.$$

Since $\left(\frac{5}{p}\right) = 1$, Corollary 3.2 gives

$$A + 3B + 3C + D_3 = L_4 = 7. \tag{3.5}$$

The threshold is $p > 2^4 + L_4 = 23$. Since $q \equiv 3 \pmod{5}$, quadratic reciprocity gives $\left(\frac{5}{q}\right) = -1$, so the $q + 1$ collapse gives

$$A + B = -2. \quad (3.6)$$

Thus $A = -2 - B$, and $A \geq -2$ implies $B \leq 0$. Substitution into (3.5) yields

$$2B + 3C + D_3 = 9,$$

but the left side is at most $0 + 6 + 2 = 8$. $\square$

For $u = 1$, this contains the gap-two situation $p, p+2$ prime. In particular $44 = 41 + 3 = 43 + 1$, while $74 = 71 + 3 = 73 + 1$ and is also detected independently by Theorem 3.3.

3.3 First representatives of the four architectures

The first appearances are not accidental consequences of the computed data; inside the range-exact low-dimensional class below they follow from the arithmetic hypotheses.

Proposition 3.6 (First representatives). *Within the range-exact low-dimensional framework of Section 8:*

- (i) 38 is the smallest positive index carrying a Type-I regular endpoint obstruction;
- (ii) 44 is the smallest index carrying a Type-II common-scale $(1, -)/(3, +)$ obstruction;
- (iii) 26 is the smallest index carrying a Type-III ramified $5^e + 1$ selector obstruction;
- (iv) 31250 is the smallest index carrying a Type-IV ramified $2 \cdot 5^e$ selector obstruction.

Proof. For (i), suppose $n < 38$ admits a regular endpoint certificate obtained by range-exact one-row-preserving descent from an outer remainder r . In the most permissive negative sign, (3.1-) requires

$$p > 2^{r+1} + L_{r-1}.$$

Already for $r = 4$ this gives $p > 36$, hence $n = p^u + r \geq 41$. Thus $r \leq 3$. A proper descent $2 \rightarrow 1$ uses the prime 2, outside the odd-prime class. A descent $3 \rightarrow 1$ would use modulus 3, but a row-1 target in $[-4, 4]$ is not range-exact modulo 3: for the positive target 3 the integers $-3, 0, 3$ are congruent, and for the negative target -2 the integers $-2, 1, 4$ are congruent. Hence below 38 an endpoint certificate is direct, with $r = 1$.

Two direct endpoint certificates for the same n have the same outer integer $n - 1$, hence by unique factorization the same regular prime power and the same Frobenius color. Thus the only Type-I possibility below 38 is a doubling atom against a negative direct endpoint certificate. If $n = 2p^u < 38$ with $p \geq 11$, then $u = 1$ and $p \in \{11, 13, 17\}$. The corresponding values $n - 1$ are 21, 25, 33; the first and third are not prime powers, while $25 = 5^2$ is ramified and is excluded from the regular class. At $p = 19$, $n = 38$ and $n - 1 = 37$ is prime with $37 \equiv 2 \pmod{5}$, giving the required negative color.

For (ii), write the common-scale architecture as

$$n = p^u + 3d = q^v + d.$$

If $d > 1$, the last nontrivial one-row-preserving descent into the positive terminal row $R = 3$ uses an odd prime $\ell \neq 5$. Range-exactness of

$$A + 3B + 3C + D_3 = 7, \quad |A + 3B + 3C + D_3| \leq 16,$$

requires $\ell > 23$, so $\ell \geq 29$ and therefore $d \geq 29$. Hence $n \geq 3 + 3d \geq 90$, certainly not below 44. Thus $n < 44$ forces $d = 1$. The positive terminal-3 certificate then requires $p > 23$, and $u = 1$ since $p^2 > 44$. The primes $23 < p < 41$ are 29, 31, 37. For $p = 29$, $p + 2 = 31$ has positive rather than negative mod-5 color; for $p = 31, 37$, the numbers $p + 2$ are not prime powers. The next possibility is $p = 41$, giving $n = 44$ and $q = 43$ with the required opposite colors.

For (iii), Type III requires $n = 5^e + 1$ together with the doubling atom. For $e = 1$, $n = 6$ cannot carry a regular doubling atom with $p \geq 11$. The next exponent is $e = 2$, and

$$26 = 5^2 + 1 = 2 \cdot 13 = 23 + 3,$$

which is Proposition 1.1.

For (iv), write a Type-IV realization as

$$2 \cdot 5^e = P + d = Q + 3d, \quad P = p^u, \quad Q = q^v, \quad (3.7)$$

with the common scale d , where p, q are regular odd primes. If $d > 1$, the row-3 certificate has a nontrivial final one-row-preserving descent into the negative terminal row $R = 3$. Let $\ell \neq 5$ be the prime used in that last descent. Range-exactness of

$$A + 3B + 3C + D_3 = -3, \quad |A + 3B + 3C + D_3| \leq 16,$$

requires $\ell > 19$, hence $\ell \geq 23$ and therefore $d \geq 23$. The initial remainder on the row-3 side is then $r_0 = 3d \geq 69$. At the outermost regular collapse, even the weaker negative threshold (3.1-) gives

$$q > 2^{r_0+1} + L_{r_0-1} > 2^{70}.$$

Thus $2 \cdot 5^e = n > q > 2^{70}$, which forces $e \geq 30$. Consequently every Type-IV realization with $e < 30$ must have $d = 1$.

For $d = 1$ one needs

$$P = 2 \cdot 5^e - 1, \quad Q = 2 \cdot 5^e - 3$$

to be regular prime powers with range-exact base primes. For $e = 1$ the row-1 base is 3 in $9 = 3^2$, so the positive row-1 lift is not range-exact; for $e = 2$ the corresponding base is 7 in $49 = 7^2$, again failing the strict threshold $p > 7$. For $e = 3$, $249 = 3 \cdot 83$ and $247 = 13 \cdot 19$; for $e = 4$, $1247 = 29 \cdot 43$; for $e = 5$, $6249 = 3 \cdot 2083$. Hence none of $e = 1, \dots, 5$ gives Type IV. For $e = 6$,

$$2 \cdot 5^6 - 1 = 31249, \quad 2 \cdot 5^6 - 3 = 31247,$$

and both numbers are prime. The direct row-1 and row-3 lifts are range-exact since $31249 > 7$ and $31247 > 19$. Section 6.3 therefore gives a Type-IV obstruction at $n = 31250$. $\square$

Remark 3.7. The statements concern the range-exact low-dimensional certificate class, not unrestricted classifications of all conceivable modular arguments. Proposition 3.6 supplies a first representative for each of the four architectures without using exhaustive search as proof.

4 Local signatures: the common mechanism

Informally, a *signature* is the set of boundary values compatible with one exact local collapse. It records only the few coordinates that another collapse may also constrain, and forgets the local variables that remain free.

Definition 4.1. Given an exact local equation involving paired variables $Y_j \in \{-2, -1, 0, 1, 2\}$, the local signature on selected global variables $X_1, \dots, X_h$ is the set of all tuples $(X_1, \dots, X_h)$ compatible with that local equation.

Theorem 4.2 (Local incompatibility). *Suppose two exact local reductions of the same index n constrain the same selected global variables and have signatures Σ_1, Σ_2 . If $\Sigma_1 \cap \Sigma_2 = \emptyset$, then $a(n) = 0$.*

Proof. Any global representation determines one tuple of values for the common variables, and that tuple would have to lie in both signatures. $\square$

Theorem 3.3 is one-dimensional: $\{1\}$ conflicts with $\{-2, -1, 0\}$. Theorem 3.5 is genuinely two-dimensional.

4.1 Terminal rows $R \leq 4$

For

$$\sum_{j=0}^R \binom{R}{j} Y_j = F_R^\sigma,$$

project to $(A, B) = (Y_0, Y_1)$. The calculation is elementary but worth seeing once. For $R = 3$ and positive sign,

$$A + 3B + 3C + D_3 = 7,$$

so

$$3C + D_3 = 7 - (A + 3B).$$

Because $C, D_3 \in [-2, 2] \cap \mathbb{Z}$, the expression $3C + D_3$ assumes every integer from -8 to 8 . Hence compatibility is equivalent to

$$7 - (A + 3B) \leq 8,$$

i.e.

$$A + 3B \geq -1.$$

The other rows are obtained by the same finite interval calculation. These inequalities should be read geometrically: each one describes the admissible region of boundary values (A, B) left by that local collapse. A contradiction occurs when two genuinely matching local regions have empty intersection.

R	sign	condition on $(A, B) \in [-2, 2]^2 \cap \mathbb{Z}^2$
1	+	$A + B = 3$
1	-	$A + B = -2$
2	+	$A + 2B \geq 2$
2	-	$-3 \leq A + 2B \leq 1$
3	+	$A + 3B \geq -1$
3	-	$A + 3B \leq 5$
4	+	full square
4	-	full square

Among the six nonfull regular signatures in the table, the five pairwise disjoint unordered pairs are

$$(1, +)/(1, -), (1, +)/(2, -), (1, -)/(2, +), (1, -)/(3, +), (2, +)/(2, -). \quad (4.1)$$

The doubling atom will add one pairwise and one genuinely three-way primitive selector; these are included explicitly in Section 8.

5 Nested digital descent and marked transport

A local collapse may leave a row that is still large but sparse in another prime base. The next theorem shows that a one-row-preserving descent is rigid.

Theorem 5.1 (One-row-preserving normal form). *Let ℓ be prime. Suppose Lucas reduction of row R modulo ℓ leaves exactly the equally spaced positions*

$$0, h, 2h, \dots, dh, \quad 1 \leq d < \ell,$$

with weights $\binom{d}{0}, \dots, \binom{d}{d}$. Then

$$R = d\ell^t, \quad h = \ell^t$$

for some $t \geq 1$. Conversely, every such R has this reduction.

Proof. Write R in base ℓ and let t be its least nonzero digit position. The smallest positive visible Lucas index is ℓ^t , so $h = \ell^t$. Any additional nonzero digit at a higher position $s > t$ would make $\ell^s = \ell^{s-t}h$ visible, with coefficient index at least $\ell > d$, contradiction. Hence R has one nonzero digit, say d , at position t . The converse is Lucas' theorem. $\square$

Thus every one-row-preserving chain has the form

$$r_{i-1} = r_i \ell_i^{t_i}. \quad (5.1)$$

If it starts from $n = p^u + r_0$ and ends at row R , define the scale

$$d = \prod_i \ell_i^{t_i}, \quad r_0 = Rd.$$

Under range-exact lifting at every stage, the terminal equation is

$$\sum_{j=0}^R \binom{R}{j} Z_j = F_R^\Xi, \quad Z_j = \varepsilon_{jd} + \varepsilon_{n-jd}, \quad (5.2)$$

where

$$\Xi = \left(\frac{5}{p}\right)^u \prod_i \left(\frac{5}{\ell_i}\right)^{t_i}. \quad (5.3)$$

Nested descent therefore preserves the small template and rescales its marked coordinates.

Lemma 5.2 (Scale uniqueness). *For every regular certificate in the class,*

$$n = p^u + Rd > 2Rd.$$

Hence each marked position jd with $1 \leq j \leq R$ lies strictly below $n/2$. In particular, if two boundary pair variables $\{d_1, n - d_1\}$ and $\{d_2, n - d_2\}$ genuinely coincide, then $d_1 = d_2$.

Proof. The initial sparse collapse has $Rd = r_0 < p$, hence $p^u \geq p > Rd$. Thus $n = p^u + Rd > 2Rd$. The rest follows because d_i is the unique smaller member of its complement pair. $\square$

5.1 The mod-5 color law

For $5 \nmid m$, write $\chi_5(m) = \left(\frac{m}{5}\right)$. Since $5 \equiv 1 \pmod{4}$, quadratic reciprocity gives $\left(\frac{5}{\ell}\right) = \left(\frac{\ell}{5}\right)$ for every odd prime $\ell \neq 5$. Hence

$$\Xi = \chi_5(d(n - Rd)). \quad (5.4)$$

For endpoint certificates $R = 1$ this becomes

$$\Xi = \chi_5(r(n - r)). \quad (5.5)$$

Thus the endpoint color is independent of the particular nested factorization of r .

If $5 \nmid n$ and $x \equiv rn^{-1} \pmod{5}$, then for a regular split $x \in \{2, 3, 4\}$ and

$$\Xi = \chi_5(x(1 - x)).$$

Therefore

$$r \equiv 3n \pmod{5} \Rightarrow +, \quad r \equiv 2n \text{ or } 4n \pmod{5} \Rightarrow -. \quad (5.6)$$

If $5 \mid n$, every regular endpoint split has

$$\Xi = \chi_5(-r^2) = +1, \quad (5.7)$$

so regular endpoint color conflicts are impossible at multiples of 5.

For the primitive common-scale pair $R = 1$ and $R = 3$, the required orientation $(-, +)$ is, when $5 \nmid n$, equivalent to

$$d \equiv -n \pmod{5}. \quad (5.8)$$

When $5 \mid n$, the same formula (5.4) instead forces the orientation $(1, +)$ and $(3, -)$.

6 The ramified prime 5

The discriminant prime 5 behaves differently: $x^2 - x - 1$ has a double root modulo 5. Nevertheless, the Lucas sequence has period 4 modulo 5, and $5^i \equiv 1 \pmod{4}$, so

$$L_n \equiv L_{s_5(n)} \pmod{5}. \quad (6.1)$$

Together with Lucas' theorem,

$$\sum_{0 \leq b_i \leq a_i} \left(\prod_i \binom{a_i}{b_i} \right) \varepsilon_{\sum_i b_i 5^i} \equiv L_{s_5(n)} \pmod{5}. \quad (6.2)$$

6.1 Ramified saturation and the complete low-dimensional list

Every paired variable $Y \in \{-2, -1, 0, 1, 2\}$ is a complete residue system modulo 5. Hence, after projecting to at most two marked complement-pair variables, any further free pair makes the projection full. Central unpaired coefficients must be treated separately.

If

$$N_5(n) = \prod_i (a_i + 1)$$

is the number of Lucas-surviving positions, a nonfull signature in dimension at most two can occur only for $N_5(n) \leq 5$. Apart from $n = 0$, this gives exactly

$$5^e, \quad 2 \cdot 5^e, \quad 3 \cdot 5^e, \quad 4 \cdot 5^e, \quad 5^a + 5^b \ (a < b). \quad (6.3)$$

Their relevant signatures are

$$n = 5^e : \quad A = 1, \quad (1)$$

$$n = 2 \cdot 5^e : \quad A \in \{-2, 0, 1\}, \quad (2)$$

$$n = 3 \cdot 5^e : \quad A + 3B \equiv 4 \pmod{5}, \quad (3)$$

$$n = 4 \cdot 5^e : \quad A - B \equiv 1, 2, \text{ or } 3 \pmod{5}, \quad (4)$$

$$n = 5^a + 5^b : \quad A + B \equiv 3 \pmod{5}. \quad (6.4)$$

For $n = 5^e + 1$, the last signature consists exactly of

$$(1, 2), (2, 1), (-2, 0), (-1, -1), (0, -2), \quad (6.5)$$

the union of the regular endpoint lines $A + B = 3$ and $A + B = -2$ inside the box.

6.2 Ramified selector I: the architecture of 26

Suppose

$$n = 5^e + 1 = 2p^u = q^v + 3, \quad (6.6)$$

with a valid doubling atom and a range-exact negative terminal-3 certificate on the q side. The doubling atom gives $A = 1$, so (6.5) forces $(A, B) = (1, 2)$. The negative row-3 signature is $A + 3B \leq 5$, but the selected point has $A + 3B = 7$. Hence $a(n) = 0$. Proposition 1.1 is the first instance.

6.3 Ramified selector II: multiples of 5

For $n = 2 \cdot 5^e$ with $e \geq 1$, the ramified endpoint set is $A \in \{-2, 0, 1\}$. Suppose there are two range-exact regular common-scale certificates

$$n = p^u + d = q^v + 3d, \quad (6.7)$$

with regular odd primes. Since $5 \mid n$ and $5 \nmid d$, the color law fixes the terminal signs as $(1, +)$ and $(3, -)$. Their signatures

$$A + B = 3, \quad A + 3B \leq 5$$

intersect only at $(A, B) = (2, 1)$, excluded by the ramified endpoint set. Thus any arithmetic realization satisfying the stated certificate hypotheses gives $a(n) = 0$.

Corollary 6.1 (Direct Type-IV prime-pair family). *Let $e \geq 3$. If both*

$$p = 2 \cdot 5^e - 1, \quad q = 2 \cdot 5^e - 3$$

are prime, then

$$a(2 \cdot 5^e) = 0.$$

Proof. Take $d = 1$ in (6.7). Then $p > 7$ and $q > 19$, so the positive row-1 and negative row-3 collapses are range-exact. The preceding selector argument applies. $\square$

The Type-IV architecture is therefore not an artifact of the finite selector census. Its significance is stronger than the existence of one more zero: it shows that the *last primitive architecture surviving the later classification is actually realized arithmetically*. Type IV exists, but within the class its first realization is unexpectedly rare; Proposition 3.6 rules out every smaller realization in the class.

Proposition 6.2 (First Type-IV realization).

$$a(31250) = 0,$$

and the obstruction is Type IV. Thus the last previously unrealized primitive architecture actually occurs.

Proof. Here

$$31250 = 2 \cdot 5^6 = 31249 + 1 = 31247 + 3.$$

Both 31249 and 31247 are prime; since their square roots are below 177, this is verified by trial division through the primes at most 173. Set

$$A = \varepsilon_0 + \varepsilon_{31250}, \quad B = \varepsilon_1 + \varepsilon_{31249}, \quad C = \varepsilon_2 + \varepsilon_{31248}, \quad D_3 = \varepsilon_3 + \varepsilon_{31247}.$$

Modulo 31249, the positive row-1 collapse is range-exact and gives

$$A + B = 3. \tag{6.8}$$

Modulo 31247, the negative row-3 collapse is range-exact and gives

$$A + 3B + 3C + D_3 = -3,$$

so $A + 3B \leq 5$. Together with (6.8) this forces $(A, B) = (2, 1)$. On the other hand, modulo 5 the base-5 form $31250 = 2 \cdot 5^6$ gives the ramified endpoint constraint

$$A \in \{-2, 0, 1\},$$

contradicting $A = 2$. □

The prime-pair condition in the corollary is only a sufficient arithmetic realization of Type IV; no infinitude claim is made.

7 Why primitive obstructions do not proliferate

This section supplies structural motivation, not a missing step in the completeness proof of Section 8. More digital blocks eventually create flexibility rather than rigidity.

Suppose a digital collapse has k independent binary blocks. Pairing complementary corners gives

$$C = 2^{k-1}$$

variables $Y_1, \dots, Y_C \in [-2, 2] \cap \mathbb{Z}$ satisfying

$$Y_1 + \dots + Y_C = V. \tag{7.1}$$

For h marked variables with prescribed values $y_1, \dots, y_h$, extension to the remaining $C - h$ variables is possible exactly when

$$\left| V - \sum_{i=1}^h y_i \right| \leq 2(C - h). \tag{7.2}$$

Consequently

$$|V| + 4h \leq 2^k \implies \Sigma_h = [-2, 2]^h \cap \mathbb{Z}^h. \quad (7.3)$$

In the Lucas setting $|V| \leq L_k$, so for $k \geq 3$ the simple sufficient condition

$$h \leq 2^{k-3} \quad (7.4)$$

forces full signature.

Thus a fixed low-dimensional projection cannot remain rigid once sufficiently many independent blocks survive. This saturation principle helps explain the rarity of low-dimensional obstructions. The classification below, however, relies only on the explicitly regular and ramified selector censuses and does not assume a general classification of multiblock matching.

8 Classification in a low-dimensional certificate class

The theorem in this section is a completeness theorem for the certificate class defined below. It is not a classification of all zeros of A399744 or of all possible modular obstructions. None of the preceding zero families or explicit nonrepresentation proofs depends on this classification.

Remark 8.1 (Why this certificate class is a natural first closure). The restrictions in Definition 8.3 are not chosen to force four answers; each records a feature already used in the preceding hand proofs.

- The split $n = p^u + r_0$ is the basic sparse geometry behind the exact small-row collapses of Section 3.
- One-row-preserving descent is singled out because Theorem 5.1 proves that these are exactly the nested descents that preserve a Pascal-row template and transport marked coordinates without changing their meaning.
- Range-exactness turns a modular congruence into an honest integer equation and hence into a finite signature; without it, carry ambiguity remains.
- One or two marked coordinates are the first dimensions in which the endpoint, weighted-row, and selector mechanisms of Sections 3–6 all appear, while remaining completely checkable by hand. Section 7 gives a structural reason to expect increasing flexibility as more independent blocks survive.
- The cutoff $R \leq 4$ is an explicit class boundary: rows 1, 2, 3 already contain every nonfull regular signature used by the paper, while row 4 is full in the boundary projection of Section 4.1. No claim is made that larger terminal rows are irrelevant.

Thus Definition 8.3 should be read as a natural first low-dimensional closure of the mechanisms developed earlier, not as a model for all possible obstructions.

Remark 8.2 (The boundary is visible in the data). The class below is not chosen to contain every known zero. The exact zero 113 is odd, while every realization of Types I–IV is even. Hence 113 necessarily requires a mechanism outside the four-type low-dimensional system. Section 9 exhibits such a mechanism explicitly: a two-digit tensor collapse together with a conditionally exact row-6 constraint. This gives an external check on the scope: Theorem 8.8 is a classification internal to a naturally bounded proof system, not a relabeling of the observed zero set.

Classification strategy. The proof has three steps. (1) Enumerate the primitive local conflicts in the finite one- and two-dimensional signature system. (2) Remove candidates that cannot occur globally because their scales, parities, prime-power decompositions, or Frobenius colors cannot match. (3) The surviving possibilities are exactly four architectures.

Before the formal census, it is useful to keep the four survivors in mind as stories:

Type	Informal description
I	one endpoint coordinate is forced in two incompatible ways
II	two admissible boundary regions do not intersect
III	a ramified selector chooses a point later excluded by a regular certificate
IV	the last remaining regular intersection point is removed by the ramified endpoint set

Definition 8.3 (The low-dimensional certificate class). An obstruction belongs to this class if it is built from the following ingredients.

- (i) A regular certificate starts from $n = p^u + r_0$ with $p \neq 5$ an odd prime and $1 \leq r_0 < p$.
- (ii) Every nested reduction is one-row-preserving as in Theorem 5.1; every descent prime ℓ is odd and different from 5.
- (iii) Every congruence-to-equality step used by the certificate is range-exact.
- (iv) The one-dimensional regular projection is the endpoint variable $A = Z_0$. The two-dimensional regular projection is the boundary pair $(A, B) = (Z_0, Z_1)$, with terminal row $R \leq 4$.
- (v) Ramified base-5 certificates are allowed whenever their projected signature on at most two marked complement-pair variables is nonfull; the complete list is (6.3)–(6.4).
- (vi) Local signatures are compared globally only when their marked complement-pair variables genuinely refer to the same global position pairs.

A primitive obstruction is a finite collection of such certificates with empty common signature intersection but with every proper subcollection having nonempty intersection.

We now make the finite completeness argument explicit.

Lemma 8.4 (Regular selector census). *For the six nonfull regular boundary signatures in the table of Section 4.1, together with the doubling atom $D = \{A = 1\}$, the minimal empty intersections are exactly*

$$\begin{aligned}
&(1, +)/(1, -), \quad (1, +)/(2, -), \quad (1, -)/(2, +), \\
&(1, -)/(3, +), \quad (2, +)/(2, -), \quad (1, -)/D, \\
&(1, +)/(3, -)/D.
\end{aligned} \tag{8.1}$$

No further primitive three- or higher-certificate intersection occurs in this finite boundary system.

Proof. The five disjoint regular pairs are (4.1). Intersecting each nonfull signature with the vertical set

$$D = \{(1, B) : -2 \leq B \leq 2\}$$

adds exactly the pair $(1, -)/D$: the negative endpoint line would require $B = -3$. Among collections whose proper subcollections all meet, the only new triple is $(1, +)/(3, -)/D$: the first two signatures meet only at $(2, 1)$, and D removes that point. Direct substitution in the six displayed inequalities of Section 4.1 shows that every other empty intersection already contains one of these seven as a proper empty subcollection. $\square$

Lemma 8.5 (Global filtering of the regular census). *Inside Definition 8.3, the globally admissible survivors of Lemma 8.4 are exactly the Type-I endpoint conflict and the Type-II pair $(1, -)/(3, +)$.*

Proof. For $(1, +)/(1, -)$ and $(2, +)/(2, -)$, genuine matching and Lemma 5.2 give the same scale and therefore the same outer prime-power split; the Frobenius color cannot be opposite.

For the $1/2$ pairs, a common odd scale d gives $n = P + d$ on the row-1 side and $n = Q + 2d$ on the row-2 side, where P, Q are odd regular prime powers. The first expression is even and the second odd, impossible.

The pair $(1, -)/D$ is precisely the rigid doubling subcase of Type I. More generally, one-dimensional endpoint signatures have only the two colors

$$S = \{1, 2\}, \quad O = \{-2, -1, 0\};$$

their conflict is Type I.

The pair $(1, -)/(3, +)$ is Type II.

Finally consider $(1, +)/(3, -)/D$. For $5 \nmid n$, the color law (5.4) never permits the common-scale orientation $(1, +), (3, -)$: the admissible common-scale row-1/row-3 orientations are $(+, +)$ or $(-, +)$. For $5 \mid n$, formula (5.4) does give $(1, +), (3, -)$, but the doubling atom would require $n = 2P$ with a regular prime power P , hence $5 \nmid n$. Thus the triple is globally sterile. $\square$

Lemma 8.6 (Ramified mixed-selector census). *Take one ramified signature from (6.4), any collection of the six nonfull regular boundary signatures, and optionally the doubling atom D . The primitive empty mixed intersections are exactly the following eleven local candidates:*

$$\begin{array}{l|l} 5^e & (1, -), (1, +)/(3, -) \\ 2 \cdot 5^e & (1, +)/(3, -) \\ 3 \cdot 5^e & (1, +), (1, -)/(2, -), (2, -)/D \\ 4 \cdot 5^e & (1, +)/D, (2, +)/D \\ 5^a + 5^b & (2, -)/(3, +), (2, -)/D, (3, -)/D \end{array} \quad (8.2)$$

Here each row means that the listed regular signatures, together with that ramified signature, form a primitive empty intersection.

Proof. Each ramified signature in (6.4) is an explicit subset of the 25-point box $[-2, 2]^2 \cap \mathbb{Z}^2$, and each regular signature is given by one of the six inequalities/equalities in Section 4.1. Substitution gives the list (8.2). For transparency, the key one-point intersections are the same ones already used in Sections 6.2–6.3: $(1, +)$ and $(3, -)$ meet at $(2, 1)$, while the sum-type ramified signature $A + B \equiv 3 \pmod{5}$ contains exactly the five points in (6.5). Checking each displayed candidate after deleting any one constituent shows primitivity; every other empty mixed intersection contains one of the displayed candidates as a proper empty subcollection. $\square$

Lemma 8.7 (Global filtering of the ramified census). *Under the matching rules of Definition 8.3, the only globally admissible survivors of (8.2) are Type III and Type IV.*

Proof. The numbers 5^e and $3 \cdot 5^e$ are odd. Any regular row-1 certificate in the class has $n = P + d$ with both P and d odd, hence n even; a doubling atom also forces $n = 2P$. Thus every candidate in the first and third rows of (8.2) is sterile.

For $4 \cdot 5^e$, both candidates contain D . But D would require $4 \cdot 5^e = 2P$, so $P = 2 \cdot 5^e$, not an odd regular prime power. Hence this row is sterile.

For $5^a + 5^b$ with $a < b$, the number is even. Any regular row-2 certificate has $n = P + 2d$ with P, d odd, hence n odd, eliminating the first two candidates in the last row. The remaining candidate is $(3, -)/D$. If $a > 0$, the doubling atom would require

$$P = \frac{5^a + 5^b}{2} = 5^a \frac{1 + 5^{b-a}}{2},$$

which has the factor 5 and a coprime factor > 1 , so it is not a regular prime power. Hence $a = 0$, giving $n = 5^e + 1$. Genuine matching of the second ramified pair with the row-3 boundary variable then forces scale $d = 1$. This is exactly Type III.

For $2 \cdot 5^e$, the sole primitive candidate is $(1, +)/(3, -)$, and Section 6.3 shows that its unique regular intersection $(2, 1)$ is excluded by the ramified endpoint set. This is exactly Type IV. $\square$

Theorem 8.8 (Low-dimensional obstruction classification). *Within the class of Definition 8.3, every globally admissible primitive obstruction is, up to exchange of certificates and common digital scaling where applicable, of one of the following four structural types:*

Type I. *A regular endpoint color conflict S/O , with the doubling atom $A = 1$ as the rigid subcase D/O .*

Type II. *The regular weighted-row conflict $(R, \sigma) = (1, -)$ against $(3, +)$.*

Type III. *The ramified selector $n = 5^e + 1$, where a doubling atom selects $(A, B) = (1, 2)$ from the base-5 signature and a negative terminal-3 certificate excludes it.*

Type IV. *The ramified selector $n = 2 \cdot 5^e$, where the base-5 endpoint set excludes the unique intersection $(2, 1)$ of a regular $(1, +)$ and $(3, -)$ common-scale pair.*

Proof. Every regular primitive obstruction in the low-dimensional boundary class occurs in the finite census of Lemma 8.4; Lemma 8.5 reduces those possibilities exactly to Types I and II. Every primitive obstruction involving one ramified nonfull signature occurs in Lemma 8.6; Lemma 8.7 reduces those possibilities exactly to Types III and IV. Scale uniqueness, Lemma 5.2, guarantees that each common boundary variable used in these comparisons corresponds to the same smaller global position, so no complement swap creates an unaccounted matching. This proves completeness inside Definition 8.3. $\square$

Remark 8.9 (Finite verifiability of the completeness claim). Theorem 8.8 has a deliberately finite target. The regular stage reduces to the seven minimal empty intersections in (8.1); the ramified stage reduces to the eleven mixed candidates in (8.2). The two global-filtering lemmas then eliminate or identify every listed candidate using only matching, parity, prime-power factorization, and the mod-5 color law. The completeness claim can therefore be checked by recomputing these two finite censuses and verifying the explicit eliminations; no unrecorded search or induction is used.

Remark 8.10 (What the theorem does not classify). No assertion is made about higher-dimensional signatures, terminal rows beyond the stated range, non-one-row-preserving reductions, non-range-exact carry-sensitive phenomena, or arbitrary modular obstruction mechanisms

outside Definition 8.3. The saturation discussion in Section 7 explains why low-dimensional rigidity often disappears as digital complexity grows, but it is not used to enlarge the scope of Theorem 8.8. The Type-V family in the next section is an explicit example beyond this boundary, not an omitted fifth case of the theorem.

9 A tensor-selector family beyond the low-dimensional class

The boundary of Definition 8.3 is genuine: the first known odd zero, 113, cannot arise from Types I–IV. We now exhibit a reusable mechanism outside the low-dimensional class.

Why a mechanism beyond Definition 8.3 is necessary. The known zero 113 is odd, whereas every realization of Types I–IV is even. Any certificate for 113 must therefore use an ingredient excluded from the low-dimensional class. Two such ingredients appear naturally just beyond its boundary: a genuine two-digit Lucas tensor rather than a one-row-preserving terminal template, and a congruence whose exactness depends on information supplied by another local equation. The next theorem packages these features into a reusable prime-pattern family.

Theorem 9.1 (Tensor-selector family). *Let $r \geq 37$ be a prime with*

$$r \equiv 2 \pmod{5},$$

and suppose that

$$3r - 4 \quad \text{and} \quad 3r - 2$$

are prime. Then

$$a(3r + 1) = 0.$$

If, in addition, $3r + 2$ is prime, then also

$$a(3r + 2) = 0.$$

Proof. The $3r + 1$ branch. Put $n = 3r + 1$ and pair complementary coefficients by

$$Y_j = \varepsilon_j + \varepsilon_{n-j}, \quad A = Y_0, \ B = Y_1, \ C = Y_2, \ D = Y_3, \ E = Y_4, \ F = Y_5.$$

Set

$$p = 3r - 2 = n - 3, \quad q = 3r - 4 = n - 5.$$

Since $r \equiv 2 \pmod{5}$, one has $p \equiv 4 \pmod{5}$ and $q \equiv 2 \pmod{5}$. Thus the p -collapse is split and the q -collapse is nonsplit. Because $p \geq 109$ and $q \geq 107$, the two congruences are range-exact and give

$$A + 3B + 3C + D = 7, \tag{9.1}$$

$$A + 5B + 10C + 10D + 5E + F = -7. \tag{9.2}$$

The first equation is equivalent to

$$5 \leq A + 3B + 3C \leq 9. \tag{9.3}$$

Eliminating D from (9.2) gives

$$9A + 25B + 20C = 77 + 5E + F,$$

and hence

$$65 \leq 9A + 25B + 20C \leq 89. \quad (9.4)$$

The two strips (9.3)–(9.4) inside $[-2, 2]^3 \cap \mathbb{Z}^3$ leave exactly

$$(A, B, C) = (0, 1, 2), \quad (0, 2, 1), \quad (2, 2, 0). \quad (9.5)$$

For completeness, (9.4) first forces $B \geq 1$. If $B = 1$, then $C = 2$ and (9.3) forces $A = 0$. If $B = 2$, the same two inequalities leave only $(A, C) = (0, 1)$ and $(2, 0)$.

Now

$$n = 3r + 1 = (31)_r.$$

Since $r \equiv 2 \pmod{5}$, Theorem 2.1 gives nonsplit target $-L_2 = -3$. Pairing complementary surviving positions in the tensor row $(1, 1) \otimes (1, 3, 3, 1)$ gives

$$A + B + 3H + 3I \equiv -3 \pmod{r}$$

for two further paired variables $H, I \in [-2, 2] \cap \mathbb{Z}$. The left side lies in $[-16, 16]$, so $r \geq 37$ makes the congruence exact. Consequently

$$A + B \equiv 0 \pmod{3}. \quad (9.6)$$

But the three triples in (9.5) have $A + B = 1, 2, 4$, none divisible by 3. This proves $a(3r + 1) = 0$.

The $3r + 2$ branch. Reset $n = 3r + 2$ and again write $A = Y_0, B = Y_1, C = Y_2, \dots$. By hypothesis n is prime and $n \equiv 3 \pmod{5}$. The base- n collapse has nonsplit target $L_1 = 1$, so

$$A = 1. \quad (9.7)$$

Modulo $p = n - 4 = 3r - 2$ the split row-4 collapse is range-exact and gives

$$S_4 := A + 4B + 6C + 4D + E = 11. \quad (9.8)$$

Modulo $q = n - 6 = 3r - 4$ the nonsplit row-6 collapse gives

$$S_6 := A + 6B + 15C + 20D + 15E + 6F + G \equiv -11 \pmod{q}. \quad (9.9)$$

This last congruence is not range-exact by itself. However,

$$2S_6 - 5S_4 = -3A - 8B + 20D + 25E + 12F + 2G.$$

Using (9.7)–(9.8) and the box bounds on the paired variables yields

$$-41 \leq S_6 \leq 93.$$

Since $q \geq 107$, the only integer in this interval congruent to -11 modulo q is -11 itself. Hence

$$S_6 = -11. \quad (9.10)$$

Eliminating D between (9.8) and (9.10) gives

$$14B + 15C = 62 + 10E + 6F + G. \quad (9.11)$$

The right side lies in $[28, 96]$, so the box bounds first leave only

$$(B, C) = (2, 0), (2, 1), (2, 2), (1, 1), (1, 2), (0, 2).$$

Using (9.8) to determine the necessary value of $4D + E$, and then (9.11) with $|6F + G| \leq 14$, gives the following complete finite check:

(B, C)	required $6F + G$ after admissible D, E	status
$(2, 0)$	-14	survives
$(2, 1)$	-19	impossible
$(2, 2)$	16	impossible
$(1, 1)$	-33	impossible
$(1, 2)$	2	survives
$(0, 2)$	-12	survives

For the three surviving rows one may take $E = -2$; the displayed residuals are attained by $(F, G) = (-2, -2), (0, 2), (-2, 0)$, respectively, and (9.8) then gives $D = 1, -1, 0$. Hence exactly

$$(B, C) = (2, 0), (1, 2), (0, 2) \quad (9.12)$$

remain.

Finally

$$n = 3r + 2 = (32)_r.$$

The nonsplit Frobenius target is now $L_1 = 1$. Pairing complementary positions in the tensor row $(1, 2, 1) \otimes (1, 3, 3, 1)$ gives

$$A + 2B + C + 3H + 6I + 3J \equiv 1 \pmod{r}.$$

Its left side lies in $[-32, 32]$, so $r \geq 37$ again makes the congruence exact. With $A = 1$ this implies

$$2B + C \equiv 0 \pmod{3}. \quad (9.13)$$

For the three pairs in (9.12), however, $2B + C$ is congruent to $1, 1, 2$ modulo 3, respectively. This final contradiction proves $a(3r + 2) = 0$. $\square$

Remark 9.2 (Why the tensor-selector mechanism lies outside the low-dimensional class). For mnemonic purposes we call the mechanism in Theorem 9.1 *Type V*, or the *two-digit tensor-selector obstruction*. This is a structural distinction, not merely a new numerical family. Types I–IV are exhausted inside Definition 8.3 by one-row-preserving terminal templates with $R \leq 4$, autonomous range-exact lifting, and the ramified selectors at 5. The proof above instead uses the two-digit templates $(31)_r$ and $(32)_r$; the second branch also needs a row-6 congruence whose exactness is obtained only after the row-4 equation has narrowed its range. None of these ingredients belongs to the low-dimensional certificate class.

The odd example 113 makes the distinction especially sharp: every Type-I–IV realization is even, so its displayed certificate cannot be a disguised instance of Types I–IV. For even Type-V examples such as 112 or 142, no uniqueness of certificates is claimed; they could in principle admit additional proofs of another kind. Thus “Type V” names a genuinely new proof architecture, not a global partition of zero indices.

Remark 9.3 (First realizations and further certified zeros). For $r = 37$ the hypotheses give

$$112 = 3 \cdot 37 + 1,$$

and the additional prime $3r + 2 = 113$ gives the consecutive pair

$$112, 113.$$

For $r = 47$ the theorem gives 142. Thus three of the first eleven positive zero indices are manifestations of the same Type-V mechanism. Further direct instances include

$$202, 832, 1672, 2032, 2272, \dots,$$

and whenever $3r + 2$ is prime as well one obtains a consecutive companion; for example 2272, 2273 arise from $r = 757$. No infinitude claim is made for the required prime pattern.

10 Rarity, discovery, and the limits of the theorem

The classification supplies a negative explanation as well as positive constructions. A low-dimensional digital zero certificate requires several coincidences at once: sparse decompositions, common marked positions, unsaturated local signatures, and the correct mod-5 color mismatch or ramified selector. Most indices do not satisfy all of these simultaneously.

Every nontrivial zero produced by Types I–IV is even. All four Types I–IV have explicit realizations, with the first Type-IV example at 31250. Type V shows concretely how the theory can continue beyond the class boundary: it explains 112, 113, 142 and makes the first known odd zero 113 part of a parametric prime-pattern family rather than an isolated exception. Among the first eleven rigorously ordered positive zeros, nine now have hand-proof mechanisms in this paper; only 110 and 132 are deliberately left without a structural classification here.

Index	Mechanism	Proof status
26	Type III	theorem-level hand proof (Proposition 1.1)
38	Type I	family theorem; first realization in the class
44	Type II	family theorem; first realization in the class
74	Types I+II	two independent theorem-level certificates
104	Type II	family theorem (Theorem 3.5)
110	–	exact enumeration; different hand mechanism suspected, not promoted here
112	Type V	tensor-selector family (Theorem 9.1)
113	Type V	tensor-selector family; first known odd zero
132	–	exact enumeration; reusable hand architecture still open
142	Type V	tensor-selector family (Theorem 9.1)
158	Type I	family theorem (Theorem 3.3)
386	Type I	additional direct hand proof; A399745 rank unresolved
31250	Type IV	theorem-level hand proof (Proposition 6.2); rank unresolved

Table 1: Known zeros discussed in the paper, separated by mechanism and proof status. “Exact enumeration” records rigorous nonrepresentation data but is not used as a mathematical certificate in any theorem. A dash means that no reusable paper-level architecture is assigned.

The paper does not prove that every zero of A399744 has a low-dimensional digital certificate, that every nonzero $a(n)$ can be certified by these methods, that Types I–V exhaust higher-dimensional, central, tensor, or non-one-row-preserving obstructions, that the parametric arithmetic conditions are realized infinitely often, or that every useful exact argument must arise from autonomous range-exact lifting.

Remark 10.1 (The two unresolved early architectures). The remaining early zeros 110 and 132 are deliberately not forced into the I–V vocabulary. Current discovery evidence suggests that they correspond to genuinely different mechanisms rather than hidden instances of the five displayed architectures. For 110 the available certificate points to a cross/central interaction involving modulus 53 after boundary rigidity; for 132 exact enumeration proves nonrepresentation, but no comparably short reusable hand architecture has yet been extracted. We therefore refrain from naming Types VI or VII: a new type should be attached only to a transparent mechanism that recurs or supports a theorem, not merely to an isolated exact zero. Neither observation is used in a proof here.

From a discovery viewpoint, the theory nevertheless suggests a finite proof-producing search strategy for a fixed n : list sparse prime-power splits, apply digital collapse, descend one-row-preserving remainders when possible, record actual marked global positions, compute the

resulting one- or two-dimensional signatures, and intersect only genuinely matching signatures. Saturation discards high-flexibility cases early, while the mod-5 color law predicts endpoint orientation. Any obstruction discovered this way can then be rewritten as a short hand proof. Computation may assist discovery or checking, but is not part of the mathematical certificate.

11 Conclusion

A399744 begins as an extremal representation sequence, but its zero terms reveal a different problem: complete failure of signed Pascal representation. The first example, 26, already exhibits the central phenomenon. Sparse descriptions of a single index in different prime bases collapse a huge Pascal row to tiny local systems, and these systems can place incompatible requirements on the same few signs.

The regular theory is governed by Frobenius and a mod-5 two-color law; the discriminant prime 5 produces ramified selector signatures; matching determines whether local conflicts refer to the same global coefficients; and saturation explains why increasing digital complexity eventually creates flexibility instead of new low-dimensional rigidity. Within the low-dimensional certificate class of Definition 8.3, these ideas lead to a complete four-type classification. The realization at 31250 completes that picture. The tensor-selector family then shows how the same obstruction philosophy continues beyond the class: it explains 112, 113, 142 and gives a direct hand proof for the first known odd zero 113.

Equivalently, the obstruction theory developed here produces and organizes members of the zero-index sequence A399745. The exact enumeration fixes the first eleven positive ranks, and nine of those eleven indices now receive hand-proof mechanisms in this paper; the architectural status of 110 and 132 is intentionally left open.

Appendix A gives a logically independent residue-class comparison that clarifies why the Lucas target is qualitatively different from its Fibonacci analogue: Fibonacci has a uniform legal two-residue construction, whereas the canonical Lucas construction necessarily doubles the symmetry-fixed residue class. The phenomenon studied here is therefore not that Lucas numbers are usually difficult to express by signed Pascal sums, but that certain indices admit several sparse digital descriptions whose local arithmetic information is mutually incompatible.

A Cyclotomic contrast with Fibonacci

This appendix is logically independent of the obstruction theory. Residue-class binomial sums associated with Lucas-type sequences have been studied in several forms; see Gordon [2, 3]. Here the comparison serves only to isolate a simple contrast: Fibonacci has an automatic legal periodic construction, while the analogous Lucas pattern has a coefficient-saturation defect.

Let $(F_n)_{n \geq 0}$ be the Fibonacci sequence, $F_0 = 0$, $F_1 = 1$. For a residue $j \pmod{5}$, write

$$S_j(n) = \sum_{\substack{0 \leq k \leq n \\ k \equiv j \pmod{5}}} \binom{n}{k},$$

and put

$$r \equiv 3n \pmod{5}.$$

Thus $2r \equiv n \pmod{5}$, so binomial symmetry gives

$$S_{r+t}(n) = S_{r-t}(n) \quad (t \in \mathbb{Z}).$$

The five residue sums have an especially simple Fibonacci–Lucas form.

Proposition A.1 (Residue-class decomposition). *For every $n \geq 0$,*

$$S_r(n) = \frac{2^n + 2(-1)^n L_n}{5},$$

$$S_{r\pm 1}(n) = S_r(n) + (-1)^{n+1} F_{n-1}, \quad S_{r\pm 2}(n) = S_r(n) + (-1)^{n+1} F_{n+1},$$

where the second line is interpreted in the evident way for the small initial indices.

Proof. Put

$$A_n = S_r(n), \quad B_n = S_{r+1}(n) = S_{r-1}(n), \quad C_n = S_{r+2}(n) = S_{r-2}(n).$$

The cyclic Pascal recurrence $S_j(n+1) = S_j(n) + S_{j-1}(n)$ gives

$$A_{n+1} = 2C_n, \quad B_{n+1} = B_n + C_n, \quad C_{n+1} = A_n + B_n.$$

At $n = 1$ the triple is $(0, 0, 1)$. Direct substitution, using the Fibonacci–Lucas recurrences, shows that the three formulas in Proposition A.1 satisfy the same recurrence and initial values; hence they hold for all $n \geq 1$, and $n = 0$ is immediate. Equivalently, the same identities follow from the standard fifth-root-of-unity filter; the recurrence proof keeps the argument elementary. $\square$

The Fibonacci analogue of A399744 is therefore qualitatively easier. Indeed,

$$F_n = (-1)^{n+1} (S_{r+2}(n) - S_{r+1}(n)).$$

Because $S_{r+2} = S_{r-2}$ and $S_{r+1} = S_{r-1}$, one may choose one residue class from each complementary pair so as to minimize the number of used positions. If $n = 5q + t$, $0 \leq t < 5$, the resulting numbers of terms are

$$2q, \quad 2q + 1, \quad 2q + 1, \quad 2q + 2, \quad 2q + 2$$

for $t = 0, 1, 2, 3, 4$, respectively. Hence the Fibonacci target always has a signed representation using at most

$$\left\lfloor \frac{2n + 4}{5} \right\rfloor$$

entries of row n .

Lucas behaves differently. Proposition A.1 gives the formal identity

$$L_n = (-1)^{n+1} (S_{r+1}(n) + S_{r+2}(n) - 2S_r(n)).$$

The coefficient 2 on the symmetry-fixed residue class is forbidden in the signed representation problem, where each individual Pascal position may be used only with coefficient $-1, 0$, or 1 . Thus the most natural cyclotomic construction is automatically legal for Fibonacci but fails at exactly one doubled residue class for Lucas. This observation is only motivational—it is not a nonrepresentation criterion—but it explains why the Lucas problem can exhibit genuine zeros while its Fibonacci companion has a uniform periodic construction.

Remark A.2. The residue decomposition above is an integral roots-of-unity statement about the positions $k \pmod{5}$. It is distinct from the mod-5 color law of Section 5.1, which concerns the quadratic character of 5 in auxiliary prime reductions.

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